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Norian magnetostratigraphy from the Scheiblkogel section, Austria: constraint on the origin of the Antalya Nappes, Turkey

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Abstract

Late Triassic magnetostratigraphic investigations in southwestern Turkey have suggested that parts of the Antalya Nappes are constituted of a melange of blocks originally deposited either in the vicinity of the northern tip of Arabia or, more surprisingly, directly north of India, in the southern hemisphere. In order to ascertain this result more clearly, we have correlated the Turkish series with Austrian sections of corresponding age from the Northern Calcareous Alps, for which the hemisphere of deposition is beyond doubt. The new magnetostratigraphic results obtained from the Scheiblkogel section (Austria) confirm the previously suggested middle to late Norian magnetic polarity sequence and support the heterogeneous character of the Antalya Nappes.

Keywords: Northern Limestone Alps; Antalya Turkey; Norian; magnetostratigraphy; paleomagnetism

1. Introduction

Several paleomagnetic and magnetostratigraphic studies suggest a complex tectonic history of the Antalya Nappes (AN), Turkey. Although there is no geological and tectonic evidence for large differential latitudinal movement within the AN, paleomagnetic results seem to indicate that parts of the AN were deposited in the northern hemisphere and others in the southern hemisphere [1–3]. This situation, which recalls the far-travelled terranes of northwestern North America (e.g. [4]) may illustrate the hemispheric ambiguity of paleomagnetism, as discussed by Butler [5].

The Antalya Nappes are located in the south of the Taurus calcareous Axis (southwestern Turkey) [6]. They are formed by three major thrust units [7–9]. Triassic paleomagnetic and magnetostratigraphic results were obtained from two of these units, the median Antalya nappes, consisting of basin deposits, and the upper Antalya nappe system, consisting of a distal pelagic plateau, basinal limestones and carbonate platform deposits. A paleolatitude of 4.7°S ± 5.8° and a small counterclockwise rotation of about 20° were proposed by Lauer [1,2] for the Kara Dere-Calbali Dag unit (median Antalya Nappes) from the paleomagnetic study of 11 sites of late Triassic (Carnian) pillow lavas. However, the possi-

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bility could not be excluded that these data might correspond to a northern hemisphere origin with a large rotation. More recently, late Triassic magnetostratigraphic results were obtained from the upper Antalya Nappes [3,10–13]. The paleolatitudes derived from the Bolücektasi Tepe and the Kavur Tepe sections [3,10] may indicate that the two sections were deposited in different hemispheres (10.4°N ± 1.0° for Bolücektasi Tepe and 17.4°S ± 1.1° for Kavur Tepe). A large differential latitudinal movement of some 3000 km must, therefore, have taken place between these two sections since the late Triassic.

In order to decipher the origin of the Antalya Nappes, we have used a magnetostratigraphic approach which consists in correlating the investigated Turkish sections with other late Triassic sections for which the hemisphere of deposition is well known. The Northern Calcareous Alps (NCA) are a very favourable place for such a hemispheric reference because middle and late Triassic sections are widely exposed and their presence in the northern hemisphere since the upper Triassic is universally accepted (e.g. [14–17]). We previously studied the magnetostratigraphy of the Mayerling section, located in the eastern part of the NCA [11]. A very satisfactory correlation was found between the Mayerling and the Bolücektasi Tepe sections for the lower Carnian (Julian 1 zone), which clearly indicates a northern hemisphere origin for the Bolücektasi Tepe section, at the northern tip of the Arabian margin. A similar test was required for the Kavur Tepe section. For this reason we have investigated the magnetostratigraphy of the middle to upper Norian Scheiblkogel section, located in the central part of the NCA. One objective was also to test the Norian magnetic polarity time scale suggested by Gallet et al. [3].

2. Geological setting, lithostratigraphy and biostratigraphy

The Scheiblkogel section is located in the vicinity of the village of Bad Aussee (Austria), about 15 km northeast of the well known locality of Hallstatt (Fig. 1). It forms a small summit just 150 m to the east of a forest road running from Waldgraben to Scheiblkogel, on the right bank of an unnamed north–south directed creek at 980 m altitude (see Austrian topographic map ÖK 1:50.000, sheet 96 — Bad Ischl; the exact coordinates are: R 480 150, H 278 250). The well exposed locality is situated in the southern limb of a major east–west trending anticlinal structure, the 'Raschberg anticline', consisting mostly of Hallstatt limestones of middle and late Triassic age [18,19]. Tectonically, this site is part of the well known Hallstatt Nappe, a major element of the Juvavic Nappe system of the NCA (e.g. [19]).

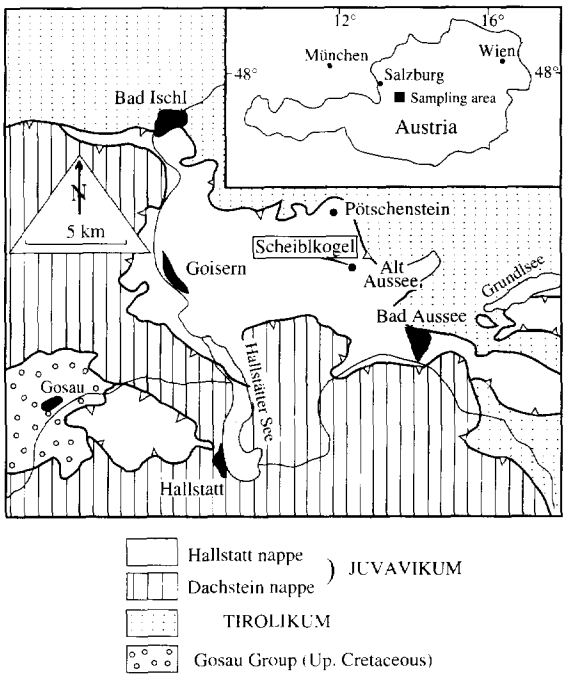


Fig. 1. Location of the Scheiblkogel section and simplified geological and structural map of the area.

The outcrop sampled is about 23 m thick, with a steep bedding attitude (strike 87°E/dip 70°). Its lithology consists of reddish and, in the topmost 2 m, greyish limestones, corresponding respectively to the Hangendrotkalk and Hangendgraukalk members of the Hallstatt formation (following Schlager [20]; see also [19,21]). The biostratigraphic zonation is based mostly on conodonts, but also on some ammonoids found in the upper part of the section. These fossils indicate an age ranging from middle to upper Norian, between the Alaunian 1 and the Sevatian 2 zones (Fig. 2).

Twenty samples were collected for biostratigraphic purposes. Within the platform conodont fauna, there is a clear dominance of the genus *Neogondolella*, whereas the biostratigraphically more diagnostic *Epigondolella* is less frequent. The zonation is based on the evolutionary lineage of *Epigondolella*, as established by Orchard [22], and is the same as in Gallet et al. [3,10]. Most major boundaries (zones or substages) have been located either precisely or to an accuracy of one bed (≈ 30 cm).

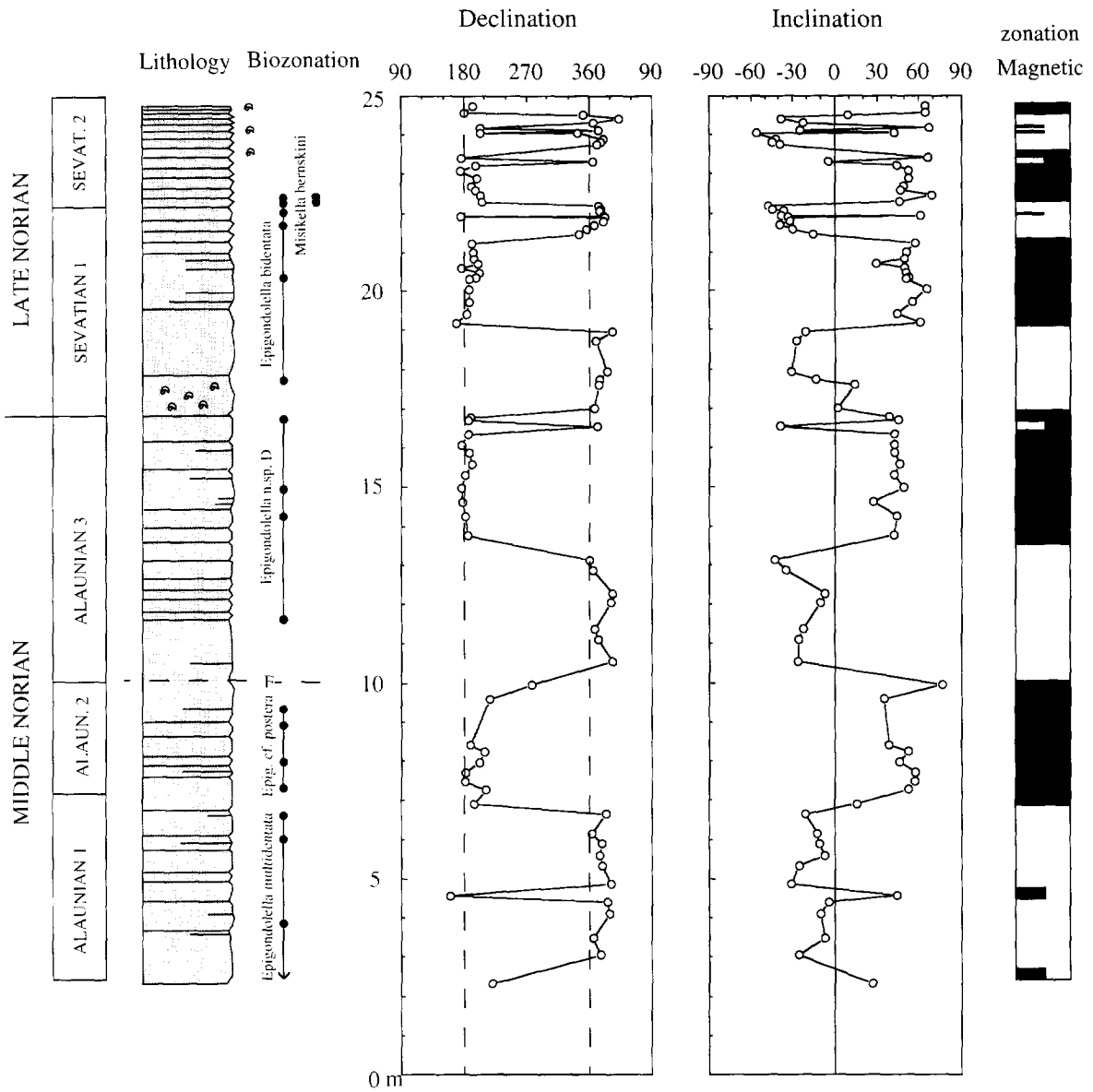


Fig. 2. Lithostratigraphy, biostratigraphy and magnetostratigraphy of the Scheiblkogel section. More than 20 magnetic polarity intervals are observed within the section studied, which consists of 23 m of greyish and reddish limestones in the Hallstatt facies. The biostratigraphic zonation is based on conodonts and also on some ammonoids for the Sevatian 2 zone.

Only the interval between 7 and 10 m is not perfectly calibrated. For this part, conodonts are not very common and the epigondolellids are dominated by juvenile to medium sized forms, whereas large (adult) specimens are almost missing. This hampers a definitive species distinction between *E. multidentata* (Alaunian 1) and *E. postera* (Alaunian 2), although a determination as *E.* cf. *postera* seems more appropriate. The inferred Alaunian 2 age for this stratigraphic interval is thus not firm but highly probable. Within the *Epigondolella* zonation, we note the absence of the *Epigondolella* n. sp. E level. It is a characteristic marker for the upper Alaunian 3 zone and, as such, is well represented in the Kavur Tepe section [3]. A small sedimentary gap between the middle and the upper Norian is, therefore, likely. Additional stratigraphic data which fit very well into the conodont zonation framework have been provided by ammonites in the upper Norian part of the section. At approximately 17 m, a 1 m thick ammonoid-rich level contains a Sevatian 1 fauna with *Metasibirites spinescens*, *Rhabdoceras suessi*, *Catenohalorites* sp. and many leiostracean ammonites (*Arcestes*, *Placites*, *Cladiscites*, *Rhacophyllites*, *Megaphyllites* and *Pinacoceras metternichi*). *Metasibirites* ranges up to 22 m, implying a Sevatian 1 age up to that level. This age is in agreement with the appearance of *Misikella hernsteini*, a conodont guide marker for the base of Sevatian 2 [23], at 22.25 m. Several fossil-rich beds, from which ammonoids can unfortunately not be extracted, occur within the topmost metre of the section, at approximately 24 m. They contain the Sevatian 2 zonal guide genus *Cochloceras*, as well as *Pinacoceras metternichi* and a diverse leiostracean fauna. The top bed of the drilled sequence shows a distinct iron oxide coated hardground known from many Hallstatt sections in upper Austria, which is often used as a first-order correlation marker. This level, called the 'Metternichi Lager' by Krystyn et al. [21], corresponds everywhere to the topmost Norian.

The whole sequence is therefore condensed. This is attributed to a sedimentation environment which is that of a pelagic plateau. The sedimentation rate is higher in the basal upper (Sevatian 1) and middle Norian. A gap seems to exist at the Alaunian 2/3 zone boundary, this corresponds to a fault which is almost parallel to the bedding plane (Fig. 2).

3. Paleomagnetic analysis of the Scheiblkogel section

Approximately 100 samples were collected during 2 field trips, in 1993 and 1994. Paleomagnetic analyses were performed using a cryogenic CTF three-axis magnetometer in the shielded paleomagnetic laboratory at the Institut de Physique du Globe in Paris.

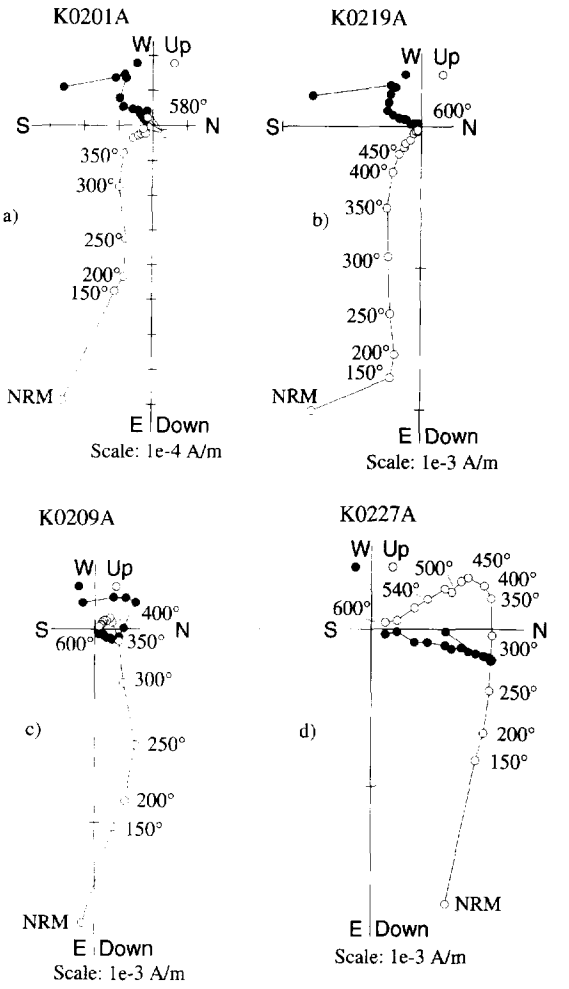


Fig. 3. Examples of orthogonal diagrams of thermal demagnetizations of whitish and reddish samples from the Scheiblkogel section. All these projections are in tectonic coordinates. Filled and open symbols represent projection on horizontal and vertical planes, respectively. (a) and (b) A normal polarity component is isolated in the high-temperature step, (a): greyish sample; (b): reddish sample. (c) and (d) A reversed polarity component is isolated in the high-temperature steps, (c): greyish sample; (d): reddish sample.

The samples were thermally demagnetized in approximately 15 steps. Magnetic susceptibility was measured after each demagnetization step in order to monitor alterations of magnetic minerals during thermal treatment. No noticeable change in the magnetic mineralogy is observed in the reddish samples during demagnetization. In contrast, the light greyish samples are often characterized by a large increase in susceptibility, after approximately 400–450°C. In some cases, spurious remagnetizations prevent a clear determination of the high unblocking temperature components. Nevertheless, most samples provided a reliable demagnetization path (Figs. 3 and 4). A first component, which, in geographic coordinates, has the direction of the present day field, was isolated below 200°C. An intermediate component, also pointing to the north in geographic coordinates, was isolated next, up to approximately 400°C (Table 1;

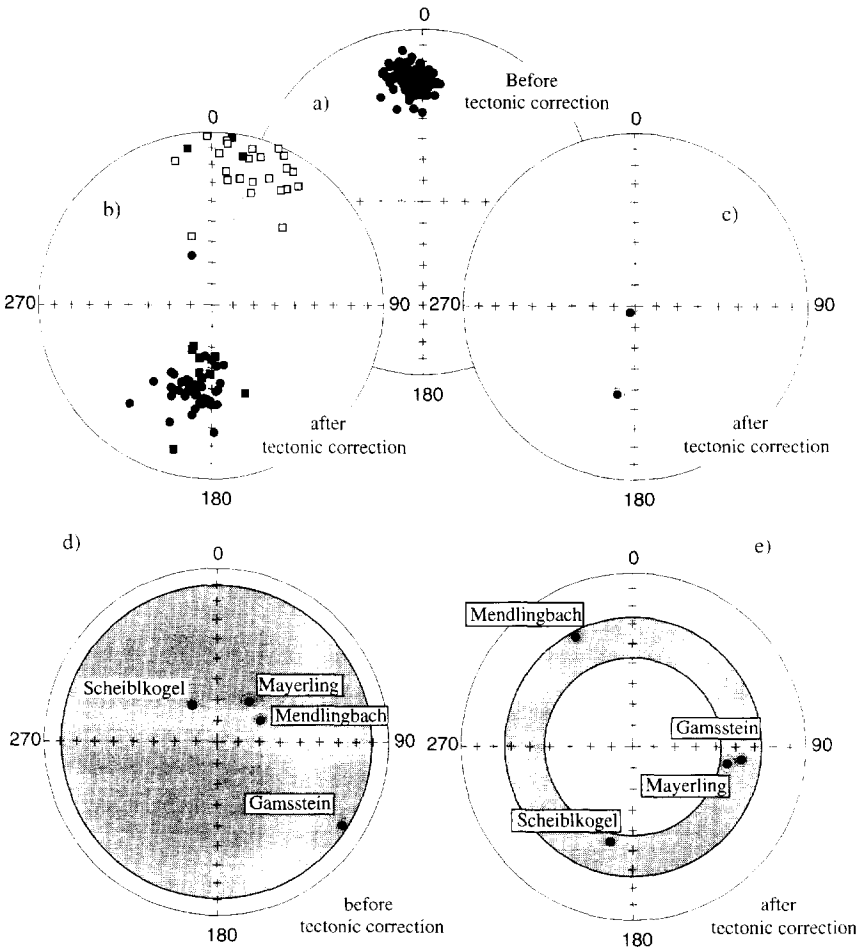


Fig. 4. Equal area projections of the directions isolated through the Scheiblkogel section. Filled and open symbols represent downward and upward directions, respectively. (a) Directions of the intermediate (200–400°C) component in geographic coordinates. (b) Directions isolated for the high unblocking temperature (450–580°C/620°C) component after bedding correction. The well defined directions are indicated by dots. The squares correspond to biased directions not considered for mean direction computations. (c) The mean normal and reversed high unblocking temperature directions are aligned along a great circle containing the mean direction of the intermediate component (in stratigraphic coordinates). (d) and (e) Mean paleomagnetic directions obtained from middle and late Triassic sections from the central and eastern NCA before (d) and after (e) bedding correction.

Table 1
Mean paleomagnetic directions of the Scheiblkogel section

Scheiblkogel (47.38°N, 13.46°E)		N samp.	Decl. before tilt correction	Incl. before tilt correction	Decl. after tilt correction	Incl. after tilt correction	a_{95}	K
intermediate component (200°-400°C)		99	355.0°	29.8°	217.2°	86.2°	1.8°	65.5
high unblocking temperature component	Normal	44	333.8°	67.4°	191.3°	46.8°	3.1°	48.6
	Reversed	49	129.6°	-72.1°	14.8°	-37.6°	2.9°	49.1
	General	93	325.4°	69.1°	193.3°	43.6°	2.2°	44.1
VGP: Lat.: -16.2° Long.: 26.0° dp: 1.7° dm: 2.7° Paleolatitude: 25.5°N								

Fig. 4a). Finally, a high unblocking temperature component, with 2 polarity states, was isolated between approximately 450°C and 580°–600°C, and sometimes 620°C in some reddish samples. The former unblocking temperatures indicate that a mineral in the magnetite family is the principal magnetic carrier in these samples, while the latter indicate that hematite may also contribute to the magnetization in the reddish lithology.

In several samples, the high-temperature component cannot be well isolated by principal component analysis (PCA), caused by a persistent overlap with the intermediate component. This is illustrated by Fig. 4b,c. In Fig. 4b the directions biased by the intermediate component are indicated by squares. For the directions with a negative inclination, the bias leads to lower inclinations. In contrast, for the directions with a positive inclination, the biased estimates generally have higher values. These biased directions have been used only to derive the magnetostratigraphy (Fig. 1) but not to calculate the mean direction (Fig. 4c). However, the mean normal and reversed directions obtained from great circles and PCA analyses [24] of apparently unbiased magnetization vectors are not exactly antipodal (Table 1; $\gamma = 9.6°$, $\gamma_c = 4.2°$ [25]), and are clearly aligned along a great circle which contains the mean direction of the intermediate direction (Fig. 4c).

4. Discussion

The characteristic component isolated in the Scheiblkogel section has a direction which points to

Table 2
Mean paleomagnetic directions obtained from different middle and late Triassic sections from the Northern Calcareous Alps [11,26]

	N samp.	Decl. before tilt correction	Incl. before tilt correction	Decl. after tilt correction	Incl. after tilt correction	a_{95}	K
Gamsstein (Ladinian)	87	123.3°	14.0°	96.8°	36.8°	2.9°	28.8
Mendlingbach (Ladinian)	37	64.0°	67.3°	332.3°	29.4°	3.0°	61.0
Mayerling (lower Carnian)	63	38.6°	65.7°	100.0°	43.6°	3.8°	23.5

the south when its inclination is down (Table 1). The declination of nearly 200° may either indicate a large rotation during nappe emplacement or deposition in the southern hemisphere. No fold test is directly available for the Scheiblkogel section. However, in addition to the presence of normal and reversed directions, other indications of a pre-folding (and early diagenetic) acquisition of magnetization are given by a comparison between several middle and late Triassic directions from the NCA (Table 2; Fig. 4d,e). Because of local rotations between the areas studied in the central and eastern parts of the NCA, this comparison should only refer to the inclinations. Together with the Scheiblkogel and Mayerling (lower Carnian [11]) mean paleomagnetic directions, we have considered the results obtained from the Mendlingbach and Gamsstein sections (Table 2; [26]). These 2 sections provide upper Ladinian (Longo-

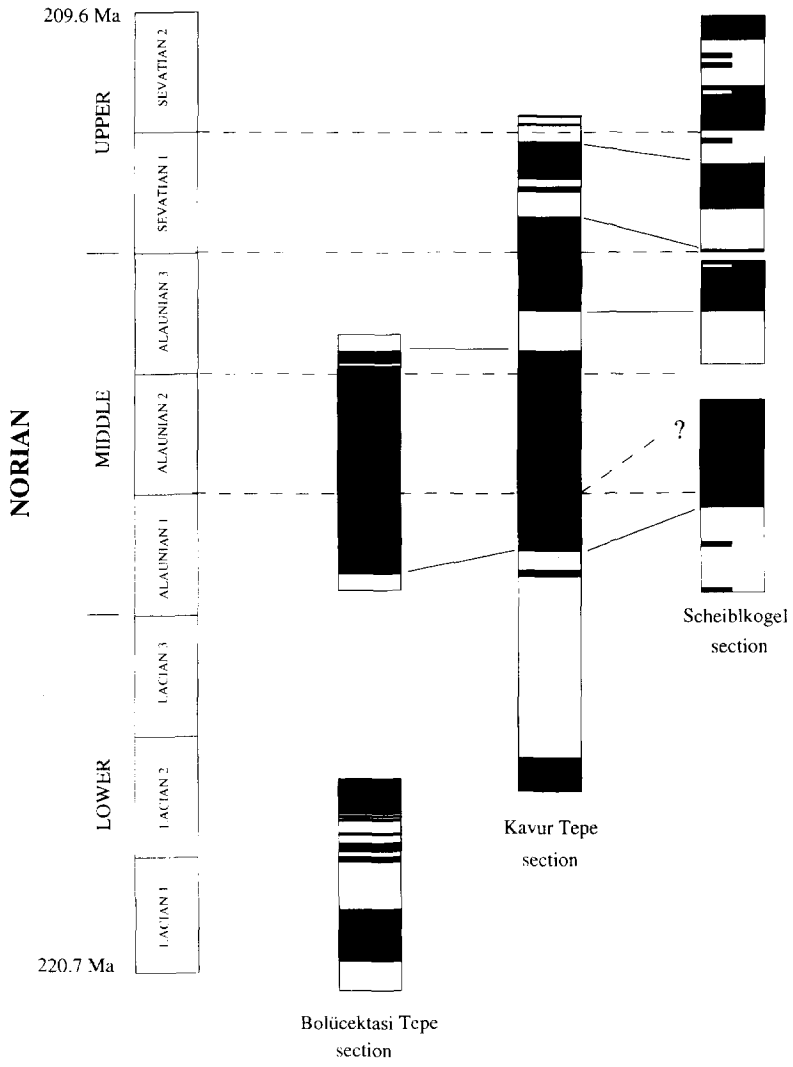


Fig. 5. Comparison between the magnetostratigraphic sequences obtained from the Bolücektasi Tepe and the Kavur Tepe sections (southwestern Turkey; [3,10]) and from the Scheiblkogel section. Age of the Norian boundaries are from [28]. A quite satisfactory correlation is obtained between the Scheiblkogel and the Kavur Tepe sections (continuous lines) if the Kavur Tepe section was deposited in the southern hemisphere.

bardian 1 and 2 zones) magnetostratigraphic sequences which can be satisfactorily correlated. When bedding corrections are made, the inclinations observed from the different sections become closer, which constitutes a (partial) positive fold test (Table 2). This good consistency in inclination is also in agreement with inclinations found from several lower Jurassic sites from the NCA (e.g. [14–17]). Moreover, two of the middle and upper Triassic sections studied, the Gamsstein and the Mayerling sections, show declinations of approximately 90° in tectonic coordinates, similar to declinations previously observed from the NCA (e.g. [16,17]). For these different reasons, deposition of the Scheiblkogel section in the southern hemisphere can be unambiguously rejected. The paleolatitude estimated from the Scheiblkogel section is 25.5°N ± 1.7°, in very good agreement with a recent paleoreconstruction of the Tethys area during the late Triassic [27]. It is worth noting that, if the Scheiblkogel section was nevertheless considered to have been deposited in the southern hemisphere, this would imply huge differential latitudinal movement within the NCA of approximately 6000 km, about twice the one suggested for the Antalya Calcareous Nappes. Finally, the rotation observed from the Scheiblkogel section is much larger than the homogeneous rotation (approximately 90°) often considered for the NCA (e.g. [16,17]). This suggests a more complex deformation of the NCA during nappe emplacement, with various rotation amplitudes (see also the direction obtained from the Ladinian Mendlingbach section; Fig. 4d,e [26]).

The magnetostratigraphy of the Scheiblkogel section is shown in Fig. 2. More than 20 magnetic intervals are observed within the section sampled. Due to the low sedimentation rate, several intervals are defined by one sample only (half bars; Fig. 2). In particular, the magnetostratigraphic sequence during the Sevatian 2 zone remains unclear and its significance is questionable. The magnetic zonation is better defined for the other biostratigraphic zones. The fault which marks the Alaunian 2/3 boundary marks a change in magnetic polarity.

The comparison with the other well dated, marine, Norian magnetostratigraphic sequences which are at present available (the Bolücektasi Tepe and Kavur Tepe sections [3,10]) is shown in Fig. 5. Because of large uncertainties existing in the Triassic absolute time scale, this comparison is made using the rough approximation of an equal duration for each of the Norian biochronologic zones. Differences in these durations would only locally contract and/or extend the magnetostratigraphic sequences, but the relative temporal (i.e. biostratigraphic) calibration of the magnetic intervals would remain identical.

Fig. 5 shows that a correlation can be proposed between the Scheiblkogel, the Bolücektasi Tepe and the Kavur Tepe sections (continuous lines), if one assumes the latter was deposited in the southern hemisphere. The new data seem, therefore, to confirm the previously suggested Norian magnetic polarity sequence [3]. The magnetostratigraphic correlations that we propose are the most likely considering the available biostratigraphic data. This correlation relies on a diachronism of limited duration for the Norian conodont species between the AN and the NCA, assuming that all the sections studied were deposited in the same intertropical band of latitude and that this part of the Tethys was an open ocean during the lower Mesozoic with no major continental barrier (e.g. [27–29]).

The magnetostratigraphic study of the Scheiblkogel section supports the southern hemisphere origin of the Kavur Tepe section, and therefore the heterogeneous origin of the AN. This conclusion is also supported by the proposed correlation between the Kavur Tepe and the non-marine Newark (eastern United States) magnetostratigraphic sequences [30]: a satisfactory correlation is also found between these 2 sections only if one assumes that the Kavur Tepe section was deposited in the southern hemisphere. The northern hemisphere origin of the Newark basin during the late Triassic is indeed beyond doubt (e.g. [30,31]). However, it must be pointed out that there is no clear correlation between the Newark basin and the Bolücektasi Tepe magnetostratigraphic sequences for the early Norian and the late Carnian.

Our study illustrates an application of magnetostratigraphy for paleogeographic reconstructions which is of particular interest in orogenic regions where large-scale rotations may occur. Only correlations between magnetostratigraphic sequences can provide a clear distinction between the occurrence of large, local rotation and sediment deposition in different hemispheres, which is referred to as the hemispheric ambiguity by Butler [5]. The implications in

terms of Tethyan kinematics of the heterogeneous origin of the Antalya nappes will be discussed elsewhere.

Acknowledgements

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